

ActInf Livestream #040.1 ~ "A free energy principle for generic quantum systems"

Livestream | 1:46:39

All right, hello and welcome everyone.

It's March 17th, 2022.

We're here in ActInfLab livestream number 40.1, discussing a free energy principle for generic quantum systems.

Welcome to the ActInfLab.

We're a participatory online lab that is communicating, learning, and practicing applied active inference. This is recorded in an archived livestream, so please provide us with feedback so we can improve our work.

All backgrounds and perspectives are welcome here, all QRF.

And video etiquette for livestream will be hopefully following.

To learn more about ActInfLab, head over to activeinference.org.

And for the rest of ActInf stream 40.1, Blue, thanks for being the facilitator.

Sure.

So, in this stream today, our goal is to learn and discuss this paper, A Free Energy Principle for Generic Quantum Systems by Chris Fields, Carl Christen, James Blaisbrook, and Michael Levin.

We have both Chris and Carl on the stream with us today.

So, we maybe want to just start off briefly with doing some introductions.

So, my name is Blue Knight.

I'm an independent researcher in New Mexico, and I will pass it off to the first author, Chris.

Hi, good morning, and thanks for attending this discussion.

This is a paper about active inference and how to place the active inference framework in a quantum information theoretic setting.

So, what I thought I should do to introduce it is just give a little bit of the motivation for doing that.

Quantum information theory is an outgrowth of quantum theory that started in the 70s, or even you could take it back to the 60s if you wanted to.

And its goal was to consider physical interactions as information exchange processes and to reformulate quantum theory in this way.

And so, it has a very natural kind of overlap with what Carl has been doing with the free energy principle in terms of reformulating physics, essentially all of physics and information and decision theoretic terms.

So why quantum theory?

Why go to this theory that's often thought of as a theory only of very tiny objects that

are moving very fast.

And the reason is that it's scale free.

And conceptually quantum theory is very simple.

And it has one core hypothesis, which is the principle of unitarity.

And the principle of unitarity is simply the idea that total information is conserved.

So quantum theory is based on a conservation law, a proposed conservation law, conservation of information.

So what going to quantum theory really forces us to do is to take seriously the project that Carl spent a lot of time on in his particular physics paper in 2019, which is to understand what we mean when we say that we've observed a system multiple times.

What we mean when we say we've interacted with the same object multiple times.

And it forces us to, or at least it encourages us to take that question seriously.

And it provides a number of tools for talking about that question.

We'll talk about that as we get into the paper.

But, and that's in a sense our motivations to take this very nice, quite abstract, but in a sense, very fundamental toolkit that's based around the idea of conservation of information, and apply it to this question of, of active inference.

So I'll let Carl pick it up from there.

Carl, thank you.

Carl, thank you.

It's a great pleasure to be here.

I have to say, I was hoping to learn more than provide a background.

So this is a great opportunity from my point of view, just to hear a conversation and participate in terms of how this health and control scale free, background free.

And that's what I think is one of the most attractive, parsimonious benefits of taking a quantum perspective on this.

It reminds me of the Aquinian desert landscape, a minimal assumption, which I'm always drawn to, and apply this to some fundamental questions, which have actually been beautifully listed for us at the top here, about things, and crucially, how things couple to other things in terms of measurement and observation and inference.

So I'm personally delighted and hungry to learn about how simple questions about the nature of things we observe and measure and make inference about can be cast from this fundamental, in this fundamental, or from this fundamental perspective of quantum information theory.

Very nice.

Thank you.

James, thanks for joining us.

Would you like to introduce yourself and maybe say a few words about the paper or pick it up from there?

Oh, hello.

You may call me Jim.

Thank you.

You want a potted biography, do you?

Anyway, I'm a retired academic emeritus professor at Eastern Illinois University and adjunct associate professor at the University of Illinois at Urbana-Champaign.

My background is global analytic geometry and some related areas.

But in the last 16 or 17 years, I've been applying mathematical techniques to cognitive neuroscience and foundations of quantum biology and, more recently, quantum information as it relates to the free energy principle.

Great.

Thank you.

Stephen, do you want to maybe introduce yourself and give a thought about the paper, something you liked or remembered or were curious about today?

Yeah, thank you.

Hi, I'm Stephen Sillit.

I'm based in Toronto.

I'm just doing a practice-based PhD into social topographies and community development.

I have got a background many years ago in chemistry.

I'm really curious about the way that this idea of something that's scale-free also ties to something that seems to be very specific on scale, i.e. where something that's highly contextual and highly bounded by degrees of freedom at a certain scale and then how something could be inferred that's across all sorts of scales.

So that's really something that I'm curious about on this particular piece of work.

Great.

Thanks.

Daniel, do you have any other, like, particular questions that you want to address or...?

Just looking forward to seeing the discussion and anyone who's watching live can ask a question.

So I'm curious about how also, like Stephen said, the specificity of a given experiment interfaces with general aspects of the formalism and many other things.

But we'll go.

Go ahead, Lou.

Well, do we want to maybe start there?

Do you guys want to speak to that scale deterministic aspect of the quantum FUP?

Well, I can say something here.

Any information theory will start with some state space, so some set of degrees of freedom that have some set of values.

And in the paper we restrict ourselves to finite state spaces because we're really interested in systems that only have access to finite energy.

And so finite computational resources.

So one starts with a state space, and in a sense that state space, when viewed from kind of the external

perspective of the theorist who's defining it, sets the scale of the problem.

Because in defining the state space, you're saying what the values are of the degrees of freedom.

And when you say, what are the possible values, you are implicitly making some assumptions about resolution.

So, and making some assumptions about measurement.

So, for example, if we think of a canonical quantum system, which is a qubit, just one, a particle with a spin.

And we say the spin can be up or down.

Then we've made an assumption about how we can measure the spin.

And the most natural mathematical framework is a three-dimensional ball of values, or sphere of values, if you normalize them.

And that's making an assumption that we can measure the spin in any one of three directions.

And so here we have imposed some possible values on this set of degrees of freedom.

And so in a sense, we've set a scale for the problem.

Similarly, if we're talking about interactions between human beings, clearly there are lots and lots of degrees of freedom that we can think of.

But we typically think about only a tiny handful of degrees of freedom that have to do, for example, with verbal communication or visual communication, olfactory communication or some other kind of communication.

And we always impose resolution restrictions.

We can say this difference is detectable and these other differences aren't detectable.

And whenever we make a decision like that, we're setting a scale.

And when we characterize the information theory as scale-free, we're just saying that the formalism doesn't change when the scale changes.

So the formalism, the mathematical description remains exactly the same, regardless of what scales we pick by choosing the possible values of the degrees of freedom that we're talking about.

We don't change anything else about the way the theory works.

So that's all that scale-free means.

It doesn't mean that we've somehow abolished the notion of scale.

That's part of our experience.

So it's not something we're going to make go away.

It's that we haven't changed the way the math works depending on the notion of scale.

And, you know, the most notorious example of that thing that we're not doing is the historical division of the world into kind of a quantum domain, a classical domain, and a cosmological domain, where you use three entirely different theories, right?

Quantum theory, classical physics, and general relativity, to talk about what's going on.

So that's not scale-free.

There you're radically altering the mathematical description to deal with different kinds of problems.

So that's what we're not doing, trying to use the same mathematical description to deal with all problems.

I think that's a really helpful description, Chris, because so many times when we think about quantum, we

think about the very, very, very small, like subatomic particles.

And so it's pretty amazing to see math that's interchangeable with my level of existence, like not just tiny little pieces of me, being used from quantum theory.

So what, is there any other math that you know of, or any other examples of math that's been interchangeable with like classical and cosmological and quantum scales ever before?

Or is this maybe the first time that's ever been done?

Well, no.

Well, no.

If you look back to classical physics in the era of Newton, or even I think the better representative is classical physics is reformulated by Laplace.

In that era, electromagnetism wasn't really understood yet.

So people were thinking about classical physics in terms of mechanical interactions and gravity.

But the idea at that point was that forces act instantaneously.

And forces act with one over r squared dependence.

That was the only force that was really understood, the force of gravity.

forces were symmetrical, right?

Newton's second law, action and reaction.

And this theory was meant to be universal.

And so if you look at the structure of that theory, as various people have pointed out,

Nicholas Geisen has written a couple of papers about this.

It's a non-local theory.

And even in Laplace's era, they realized that there was a deep issue with this theory that, you know, if a rock rolls downhill on the moon, then that immediately affects the state of everything on Earth.

In fact, immediately affects the state of everything in the universe.

So that starts to look a lot like entanglement.

That states are not locally independent.

And that's the fundamental thing that Einstein attacked about classical physics in formulating the theory of relativity,

was to insist on non-instantaneous forces, which just means forces transmitted by some carrier, which we call light.

But light is really just a carrier of information.

It could have been anything.

It could have been neutrinos or something like that.

So what Einstein did that was most revolutionary was to impose locality on physics.

And that, in a sense, divided the world into realms that, from a formal perspective, not just a practical perspective, could be treated independently.

And so it's no surprise that Einstein hated the idea of entanglement.

Because entanglement was undoing precisely what he had done to, in his mind, repair physics.

So yeah, I mean, classical physics of the time of applause was formally a scale free theory.

That's awesome.

Thank you for the clarification there.

So Einstein, Einstein broke physics.

Stephen, did you have a question?

Yeah, just one question sort of following from that is, as someone sort of thinking about how this might play out in other ways, is it useful to think about things all the way up from the smallest, and systems all the way up down from the biggest, and classical as being when we're trying to name things as systems?

So that, because often I hear people talking systems, systems, systems only, but the use of things seems to be important as well.

And I'm wondering if we should have more thinness in our way of looking at the world.

Shall I comment on that?

Yeah, okay.

Yeah, this, I think you're raising here a really interesting question.

And when we talk about things, as Carl so beautifully emphasized in his 2019 paper, we are implicitly talking about something that has an identity, and that maintains that identity through time.

And that's a deep philosophical assumption that's nearly always implicit.

I mean, people just don't talk about that assumption very much.

And when we use the word system, we're, I think, often trying to speak a lot more loosely.

But in fact, our language doesn't let us speak more loosely.

If I talk about a system, then implicitly I'm saying, I've got a particular state space.

That's what a system is.

And if I talk about a different state space, then I'm talking about a different system.

So, let's apply that to something interesting, like embryology.

I pick a set of degrees of freedom that are meant to be the degrees of freedom of the zygote.

And I then, you know, a few days later, have a set of degrees of freedom that are the degrees of freedom of some kind of early stage embryo.

But I want to say this is the same thing as the zygote.

Well, I'm now really abusing my notation, right?

In a really serious way.

By saying that these two are the same thing, even though they have different state spaces.

So, this is a, I think this is a fundamental question for any science that bases itself on the notion of a state space.

How to talk about moving between different degrees of freedom and associating those things with some kind of evolution of a kind of meta thing that we regard as a single entity.

And this is, this is one reason that, at the beginning of this paper, we, we assume that we have some single system that we divide in two.

We draw a decompositional boundary.

And that decompositional boundary is basically arbitrary, or it's completely arbitrary.

But when we're, we're doing something like embryology, we're looking at a decompositional boundary

between the thing we're interested in, the embryo, and the rest of the universe, including us.

That's getting bigger and bigger and bigger.

And so we need it, we need a way to talk about that sort of thing.

And we just, we allied that problem in this paper.

We, we talk about a fixed decomposition.

That, that divides the entire universe into two parts that we can characterize the interaction between.

But, but yes, underlying all of this is this much larger question of how we think about decompositional boundaries that are expanding.

So, with respect to some parameter that we call time, and of course that time is not anyone's observable time.

Because we're, we're trying to talk about it though, at the level of the universe as a whole.

So we, you can very quickly get into extremely deep questions.

So two things that came up there, really for me is like, okay, yeah, you're thinking about the embryo.

Suddenly that's like a system that's separate from the world.

But even like within the embryo, like where do you draw the line between the placenta and the embryo and when?

So, so like there's even some fuzzy blurring at, at the developmental scale.

And also really the notion of, of time and time irreversibility and memory and encoding was just something I found absolutely fascinating about this paper.

So, so you mentioned that the, the quantum, the free energy principle for generic quantum systems does not imply like a space time background.

Like it doesn't need a space time background to operate, but then like brought time back in, in terms of memory and encoding.

And, and I, I just wonder if, if anybody maybe wants to speak a little bit more about that or, or help us understand how and why you don't need space or time for it to work.

Hmm.

Well, I could talk about that a little bit if, if no one else wants to pick it up.

Let me go back to the first thing you said though, Blue, which was about boundaries and how hard it is, for example, to, to draw boundaries in something like embryology.

Oh, there's another deep, deep assumption that we make that the boundaries that we draw in the world are somehow ontological, right?

Objective and observer independent.

And, and, uh, quantum theory and, and many other sort of theoretical approaches remind us constantly that that's a bad assumption.

That, uh, drawing, thinking of these boundaries as ontic causes all kinds of formal problems.

So, for example, the associativity of the state space multiplication operator becomes invalid.

So you can, you have to throw out the whole vector space formalism.

Oh, so, I mean, real problems.

Oh, so, so boundaries are observer dependent in any formalism like this.

flow operators so if this flow operators have a dynamics then implicitly you've got a rate of change with

respect to something which is time but uh you know that's quite a big move um and um so putting things in terms of mappings I think is absolutely fundamental so Steve if your point was when you need to think about systemic processes or systems from the point of view of mappings usually associated some time I get the impression from um working with Chris and Jim that um you know there is an ordinal or a sequential aspect so just going back to the question is where did memory get into the game and you know that there has to be um there has to you know there can still be sequences or um you know sets of things um that has to constitute a

memory but it doesn't have to have the attribute of um of time in the sense that we would understand a flow from the point of view of a differential equation so I'd like to hear I'd like to hear sort of um Chris and Jim's uh comments on that while I'm talking though it um just this importance of um time as a construct um it does strike me that you know that takes um there's a lot of heavy lifting if we just abandon the notion of some universal or clock time in relation to what I would um from a classical perspective um read the scale free aspect of and I'm thinking here of the renormalization group so I'd be interested to know if that you know and I'm sure there is if there is a renormalization group or the apparatus of the renormalization group in quantum physics but from my perspective um that that notion of having a scale free approach is exactly um what is the uh aspiration of any um in particular the free energy principle approach um would have when trying to account for the progressive um increase in the scale of a system but preserving exactly the same dynamics the same lagrangians the same so again from a classical perspective the same um the same functional forms for the the things that are conserved so you know you know just to reiterate chris's fundamental point earlier on that um we're not talking about um a commitment to a particular scale be it very very small or very very big what we're talking about is a commitment to the same functional forms and the apparatus and the mathematics that underwrite the the dynamics if you like and that's you know at any given scale and then that calls then into question how would you get from one scale to the other um and i'm one thing that sort of um jumps out to you when you when you look at the the renormalization group in relation to um summarizing dynamics of random differential systems is that you get this time dilation or this you know you move from one time scale to another time scale um so i'm i'm coming back to the point that you know when you talk about time what scale what scale what time are you talking about are you talking about evolutionary time are you talking about um uh sort of you know the time that um the fetus is enjoying in its mother's womb or you're talking about the time scale appropriate for the um the fluxes of ions over your cell membrane um so the deeper question here is that how does one temp one time relate to another the people that i know who are interested in this issue from the point of view of the perception of time or time as a construct um usually fall back on information rates they usually fall back on the the rate of belief updating or the amount to which you have moved in some belief space technically statistical manifold from a classical point of view but as soon as you say rate you say with respect to what um and then of course you're back with oh some time which time and where you end up where you end up is well okay um i can't talk about any absolute information rate but i can certainly talk about um the number of moves i make at one scale per number of moves i make at another scale uh even we could be doing that in our

heads you know when we um very high scales um say from the point of view of a generative model in our in our heads dealing with things that unfold very slowly in relation to fast food updating much closer say to our sensory organs or um actuators um that it does make sense you know to actually talk about how much have i moved how how far have i moved in terms of an information length um in my belief updating in um at a higher level of a hierarchy or your higher scale in relation to the to the number of moons or how far i have moved

um
we've got somebody chatting away from that all good um jim do you want to mute if you're uh have background noise please
okay okay or you can just um um it'll be muted on live stream

That little response was very rambling because there are so many questions that have come up over the past 10-20 minutes. I'd like to pass back to Chris and Jim now, though, just to get their take on how comfortable they would be by taking the notions of, say, the renormalization group as applied to pathological formulations of bandedianic systems and whether that has any homolog or any currency in a truly background-free quantum formulation.

Well,
basically I think what Carl is proposing concerning the renormalization group and van der Waals theories is something to be investigated. But I'm in consonant with Carl concerning the aspect of memory from the informational point of view. And may I speak about the specifics of our construction, namely the Cone Diagram, which is a distributed...

I think I'll be there.

Hello?

Hello?

Yep.

Continue, Jim. I...

Okay, you could mute or just... Yeah.

Right.

I can't help you.

Well, I can't help you.

That's all good.

Well, I know.

Well, I know.

We're going to be looking at the cone...

Cocone?

Um, maybe just Chris or Carl, you can give a first pass.

Like, what is this being shown and also how does it relate to active inference?

Oh, hello.

Okay. Welcome back.

There you are.

Continue with the Cone.

Yes, I'm going to go with local distraction.

Um, yes.

Uh, okay.

Well, you see, at the beginning of the paper, when we discussed the um, holographic screen, which underlies the Markov blanket, you have um, these preparation, uh, measurement operations, which feed into the diagram, diagram 14, um, of which, um, the lower part emanating from D prime is, um, a, uh, memory write system.

So as the, um, the top part is picking up information from the measurement and preparation, uh, operators, um, there is a memory writes system being, um, uh, implemented, um, from the opposite, um, direction.

Um, one thing I, I would like to say about the individual classifiers that we haven't sort of, um, fully mentioned in the paper that an individual classifier, uh, can be the result of, um, an algebraic operation on, on two other, uh, uh, classifiers.

Um, these algebraic operations, uh, can be of the type such as, um, uh, concurrency, um, or, um, uh, orthocurrents.

For example, when you have a system of trains going through a system of stations, um, uh, uh, the, uh, the classifiers in question, uh, can be, um, combined to that extent and also by choice.

So I, I think the scope for what is happening informationally in diagram 14 is, uh, quite significant.

Um, originally, um, choose spaces were applied to such areas as, uh, as, as concurrency and, um, higher dimensional, uh, automata.

Um, and also, um, we, we can look at the possible, uh, possibleistic, uh, case.

And there are ways of interpreting diagram 14 in terms of the graph network too.

So, um, I know this doesn't exactly answer, uh, Carl's, uh, questions, but, um, I, I, I, I want to say that there is the mechanism there, um, for the, uh, memory, right.

So, um, this is, uh, of course, uh, influential when, uh, you, you see that, um, active inference leads to minimizing the variational, uh, free energy, uh, principle.

Uh, if you have any questions about those diagrams, uh, please ask me, uh, perhaps, uh, Chris, um, can elaborate on, uh, what I'm saying.

Uh, yeah, let me make a background comment here, uh, which in a sense goes to, uh, one reason that we're using this quantum theoretic formalism.

If we, if we think of everything in terms of, uh, a, a quantum theoretic formalism, instead of a classical formalism, then the only place that there's any classical information in the system is actually on the boundary between the two interacting components.

So, um, in that original, I, I don't know whether it's figure one or, or the, the, the very original idea in the paper is that we, we have some closed system, which we can think of as the universe as a whole, or the universe of interest as a whole, and it's in some quantum state. And if we divide it into two and we make the assumption that the joint state after the division is separable, that we've divided it in a place where the interaction of the system with itself is weak enough that we can think of the two halves independently or the two components independently, then we have this natural sense of classical information written on that boundary. And that's the sense of classical information that one encounters in the holographic principle in cosmology. And it's, it's in that sense of classical encoding that this boundary serves as a Markov blanket. But since the boundary is the only locus anywhere of classical information, in this formal description, then any classical memory has to be resident on that boundary. And this actually brings to mind a picture which was not included in this paper, but that I'm using in a subsequent paper with Mike Levin, which really is, is just thinking of this boundary using a pie chart. And as we

pointed out in this paper, another kind of classical information is the classical free energy flow that powers information processing. So this, this actually isn't the right picture, but...

And so one can immediately carve off a large chunk of the boundary and say, okay, this is just thermodynamic exchange. These bits aren't useful to anybody.

They're, they're the bits that are being used burned as power. Yeah, this is a better picture for that.

And then there's a chunk of the pie chart that we can think of as perception, and another chunk of the pie chart that we can think of as action. But then some other piece of the pie chart has to be allocated to memory.

Or there's no internal time. Right? Having an internal clock and having a memory are the same concept.

If I don't have any memory at all, then I have no awareness of the passing of time and I have no internal clock.

So when Jim talks about, in that previous Cone-Code Cone diagram, a memory-write operation, it's a memory-write operation on the boundary.

And of course, the opposite partner in the interaction. Now, my memory segment of the boundary, from my point of view,

might be my partner's perception segment of the boundary. Or might overlap with my partner's perception of the

segment of the boundary. I have no idea how my interaction partner cuts up the boundary from their point of view.

So, this is, yeah, that's what this picture sort of illustrates.

So, this question of what is the memory also becomes observer independent, which gets back to this question of time and the

relationship of time to a local clock and the relationship of a local clock to memory.

So, whenever we talk about time as observed by someone,

we're making reference to a local clock and a memory resource and, of course, an energy flow that powers that memory

resource and prevents it from decaying, uh, entropically.

So, this, I, we keep circling back to this abstraction of the universal time that parameterizes the universal dynamics.

Um, but, um, and, you know, this, this formalism, like any such formalism, requires that concept as a theoretical construct.

Um, but, um, at least it doesn't confuse it with the, the experienced time of any particular observer, which has to be represented in this very different, in this very distinct way.

Awesome. Thanks.

So, I think that is a perfect segue into a question that was asked by Franz Kutchling in the live chat.

Um, and also Mike Levin has joined us. I don't know if he's able to talk. He says he's having some internet issues.

Mike, did you want to say hello before I ask Franz's question? If you can?

You're unmuted. Um, okay. I'll ask the question and then we'll see if Mike figures out how to unmute and say hi.

Um, so Franz asks, how does the definition of a thing that maintains its identity over time scale with time scales?

Or, over time scale with time scales. There you go. How does the definition of a thing that maintains its identity over time scale with time scales?

What? I still didn't get it. For example, I would call something a thing if it persisted for some time with respect to my observational time scale. So, in other words, can quantum theory say something about how to set temporal observer limited boundaries of defining a thing?

There's a lot in there.

I want to say, so how is the definition of a thing?

Can I give one for a stop, Lou?

Yeah, yeah.

Um, we see many ways, um, some that are actually implicitly scale friendly or scale free, like the hierarchical parametric models in SPM, statistical parametric modeling.

That's separating like in a multi-scale experimental design from an experimentalist's perspective into signal and noise, including the combinatoric interactions in a classical parametric state space.

And there's also, um, the, uh, Bayesian identification of parameters, which also is just as was brought up, scale implies choosing what the state space is.

And so what statistical variance gets partitioned into the noise term or the kind of ripple term and what gets partitioned into the wave term, into the signal, into the regression slope, rather than the variance term.

These kinds of variance partitioning questions can be approached in an instantaneous sense, like descriptive statistics, even if it's about time series.

And, um, um, I think a theme that we're hearing brought up is that taking repeated measurements seriously brings in a whole host of essentially cognitive features, like as well as, uh, information, theoretic, um, perspective, dependence, all of these features that are, um, tucked under the rug.

Or, um, down projected onto in a special case.

If we just assume a synchronous universe, um, or a synchronous regression, it's a totally different scenario.

Um, but considering this quantum statistical background helps us deal with the cognitive modeling and also take the consequences of action into account because action is implied in the measurement preparation recording experimental active inference cycle.

It was just so subtle.

It was just so subtle in the paper where we said all inference in the scheme is active inference, but the paper was about quantum and the FEP.

Thanks.

Uh, could I bounce off what Daniel just said very quickly?

Sure.

Because, uh, I was also thinking about the nature of action.

I thought I was, um, so I thought what Daniel just mentioned there ties in with is, is if memory happens via there being some action and then the action defines the memory scale.

There's, um, so I'm, um, I think that makes, that, that, that, that starts to define a scale.

So I'm curious that if, if there's something different between the, the temporality of a thing maintaining its thingness and the sort of temporality of, um, actions by systems of things.

Because I think things can only act to maintain their thingness, but they don't, if they start doing things with others, they become a system.

So I suppose you've got where, where that action still starts to come into the equation or am I mistaken there?

Can it, can I address that briefly?

Um, yeah, I, I think a relevant,

the relevant fable here is the, the fable of the two drunks who are walking home from the bar leaning on each other.

Uh, and, uh, if they're lucky, they managed to keep each other upright.

Uh, if,

if, if, if we, if we think of

an observer in an environment,

and so we draw a boundary between

the observer and the environment,

and we think about the observer's interaction with its total environment,

then we've divided the boundary up,
as we were talking about earlier,
uh, into a pie chart where,
uh, you know,
a, a large fraction of the bits are fuel,
and so they're uninformative,
and there's a perception space,
and an action space,
and, and a memory space,
and so on.

Now let's think about what happens when the observer identifies a thing embedded in the environment.

So the observer has now segmented,
or sectoried,

the,
the perception part of the space,
into a part
that includes bits that represent the state of this thing,
and the rest of the perception space,
which are bits that represent the background
that the thing is embedded in.

So,
the things environment.

Um,
and so the question becomes,
what does it mean
to say that that thing
remain,
retains its thingness,
its identity over time,
for that observer?

Um,
well that means that
that observer
is able to
maintain this distinction
between
bits that communicate information about the thing,
and bits that communicate information about the thing's environment.

So this becomes a cognitive question about the observer.

So,

it's

the question,

how does a thing

maintain its identity,

is

replaced in,

in this way

by the question,

how does an observer

of a thing

maintain

the thing's identity

for

himself,

or for itself?

How does the observer

maintain the ability

to lock in on this thing

as distinct from the environment?

movement?

And,

um,

so let's,

let's go to an example,

right?

Think about motor babbling

in an infant.

So,

here's this infant,

she's lying on her back,

uh,

she's waving her arms

and her legs around,

and,

what she's doing

in this motor babbling

is
figuring out
the relationship
between
the motor system
and the visual system.

So,
among other things,
she's figuring out
that some,
that something that she sees,
some visual stimulus,
is actually part of her body.

It's,
it's a hand.

And,
that at a certain moment,
she begins to be able to control
what this hand is doing,
uh,
using visual and motor feedback.

So,
here's,
here's a living example
of an observer,
uh,
fixing the identity
of an object.

And,
what we're trying to do here,
in a sense,
is,
use this as a,
is,
is to insist on thinking about
thinghood always,
in this observer-specific way.

So,

I noticed that Carl
just raised his hand.
So,
let me turn it over
completely to him
to carry on with that,
or contest it,
or,
or give a different perspective.
No,
I don't,
I don't want to contest it.
I want to fully endorse that.
And,
if you just,
um,
look at
the free energy principle
as classically formulated,
it is exactly,
uh,
along the lines
of the,
uh,
the thingness
being an attribute
of the observer.
So,
if you just start off
with the notion
of a Markov blanket
that stipulably defines
a thingness,
how is the Markov blanket
defined?
Well,
it's defined
by the conditional

independence
between the dynamics
on the inside
versus the outside.
But,
of course,
those dynamics
are not observable
once they're on the inside
from the outside
by definition.
So,
you're immediately
imputing
some magical,
um,
observer
that could actually
recognize
that independence
before you've even
defined what a thing is.
I just wanted to,
to,
to,
you know,
um,
in an abstract
and possibly even
philosophical way,
um,
reinforce this notion
that thingness
does,
you know,
is an attribute
of something
that could possibly

observe it,
even from the
classical perspective
of the free energy
principle predicated
on the notion
of a Markov blanket
that is in turn,
uh,
defined in terms
of conditional
independencies
or disentanglement.

Um,
that,
that,
that,
that example
of the,
um,
of motor-baviling,
I think is,
you know,
again,
a lovely,
uh,
illustration
of,
um,
observation
underwriting
thingness
and in this instance,
um,
you know,
a highly anthropomorphized
thingness,
namely selfhood,

you know,
to,
to develop
the hypothesis
or the explanation
for my holographic
screen
or my,
um,
uh,
all the,
uh,
information
on my Markov blanket
that,
uh,
a,
a plausible
parsimonious explanation
is that I am a thing,
I am a self,
I am me,
is,
is actually,
you know,
takes months
if not years
to,
to fully
actually,
uh,
you know,
arrive at
as you're an infant,
slowly discovering
that I am different
from mum,
um,

and perhaps mum
is different
from the back,
her background,
um,
ultimately,
perhaps I am
something like mum,
uh,
so,
yeah,
the,
the,
these,
uh,
the emergence
of thingness,
I think,
it is,
it is quintessentially
observer-dependent,
both mathematically
and in terms
of developmental,
uh,
psychology.
great,
thank you.
Oh,
Carl,
did you want
to add up?
Sorry.
I was trying
to lower my hand.
I'm just going.
It,
it does that by

itself,
I think.
Oh,
how clever.
How lucky,
the technology.
So,
maybe I will
put,
put this back
to Carl.
Maybe you were
talking earlier
before Jim
started talking
about the rate
of information
flow and the
rate of belief
updating.
And so,
I just wonder,
I've been curious
about this actually,
um,
just in terms
of human
information,
like,
are we really
bombarded with
more information
now than we
used to be?
Um,
and is this,
like,
influencing our

rate of cultural
change,
cultural evolution
because as a
culture,
like,
we update our
beliefs more
often.

Do you think
that these two
things are
proportional,
the rate of
information flow
and the rate
of belief
updating?

Something I've
been thinking
about.

Yes,
no,
I'm sure
that's absolutely
right.

And right,
I think,
you know,
at every scale
of analysis.

So,
you could look
at it from a
point of view
of culture
and the
availability

of information
on social
media and
globalization
as a cultural
scale.

you could
also,
I think,
bring us
back to
motor babbling.

You know,
the degree
of belief
updating
an infant
has to
do is
clearly going
to be much
greater than
somebody of
my age.

So,
you know,
time will,
a year in the
life of an
infant will not
be the same
as a year
in my life.

And,
you know,
I think that
that is a
direct reflection

of the degree
of belief
updating.
Interestingly,
that question
touches upon
the separation
of time
scales,
which I think
underwrites
everything that
we're talking
about here.

So,
just to stand
back again
and go back
to some of
Chris's
early observations
about the
remit
of the
paper in
question,
it was
really
at any
given
but not
specified
scale
where we
could assume
that
we didn't
have to worry

about the
separation
of time
scales.

From the
point of view
of a
cognitive
neuroscientist,
for example,
what we're
talking about
is inference,
where inference
in this instance
would be read
as a kind
of reading
and writing
onto this
holographic
screen,
or if you're
one of the
classical
perspective,
the updating
of active
and sensory
states on
the Markov
blanket.

But under
the assumption
that other
things that
change at a
slower time

scale are
not changing
and those
other things
might be,
for example,
at a slower
time scale,
they might
be learning.
So you've
got this
notion that
the formalism
at hand,
the reading
and writing
at hand
from the
quantum
information
theoretic
perspective
could be
a very good
metaphor for
inference
over a
short time
period
where
things are
not changing
very quickly
from the
point of view
of the
things to

be learned.

So within

a few seconds

or minutes

of the life

of the

zygote,

acknowledging

that you

could apply

the same

apparatus

and the

same

mechanics

at a

slower

time scale

that

would

account

for

the

transition

from a

zygote

to a

neonate

as it

learns.

So the

difference

between

inference

and learning

I think

is a

nice

common
sensible
example of
this
separation
of time
skills.

So when
you ask
about
cultural
information
available

I think
you're
probably
talking
about

a
scale
of
learning
and
absolutely
when we
learn

we update
our
beliefs
there will
be from

a
classical
perspective

a
statistical
manifold
that

holds
those
beliefs
and you're
literally
moving
over
that
and with
respect
to
another
scale
then
the
degree
of belief
update
and the
degree
of asking
questions
the degree
of read
writing
will
certainly
be
a
function
of the
amount
of reading
and writing
that's going
on
and I can
imagine

that
somebody
who has
access to
Wikipedia
or news
or social
media
is doing
a lot
more
read
writing
than I
was
in the
1960s.
so
Carl
you made
a
fascinating
point
about
connecting
the
motor
babbling
and taking
it into
the
informational
and the
belief
updating
domain
into
the

adult
where
we're
kind
of
motor
babbling
on
social
media
in a
way
that
has
different
dynamics
than
before
and
people
often
give
the
example
of
like
well
a
year
feels
longer
for
a
seven
year
old
because
it's

a
seventh
of
their
life
but
then
for
the
80
year
old
it's
an
80th
of
their
life
so
that
piece
of
the
pie
is
smaller
versus
bigger
and
that
is
implicitly
the
fallacy
of
the
observer
independent

time
as if
there
was
a
linear
chart
that
was
being
linearly
divided
by
some
or
experiences
or
change
or
novelty
or
continue
to
explore
and
don't
right
I'll
mute
James
oh
he's
muted
on the
live stream
so
that
rate

of
belief
updating
could
set
that
internal
kairos
that
biological
time
like
that
biological
information
geometry
and
so
it
connects
the
cognitive
sciences
and
also
the
statistical
perspectives
that
have
been
discussed
to
this
time
question
which
as

also
brought
up
earlier
does
so
much
work
in
so
called
dynamical
modeling
and
it
just
makes
me
think
about
how
there
was
a
movement
from
the
observer
independent
synchronous
into
the
relativistic
which
prepared
and
almost
like

prepared
the
partition
to
exist
so
that
all
these
other
domains
like
the
holographic
principle
could
come
into
play
so
my
question
for
any
of
the
authors
is
where
does
the
QRF
come
into
play
and
how
does

this
relate
to
discussions
of
pluralism
and
polycentrality
awesome
question
Jim did
you
want to
answer
well
I wonder
if you
can bring
up a
diagram
there
in the
paper
let's
see
yes
it's
diagram
34
there
you
have
configuration
that
relates
the
QRF
to

the
coca
cone
diagrams
and
we
say
here
that
let's
see
well
here
we
focus
on
co
deployable
observers
when
this
diagram
commutes
is
interesting
of course
when the
diagram
doesn't
commute
and then
you have
a kind
of
intrinsic
context
reality
so

I
think
what
this is
saying
is
that
there
are
certain
elements
of
information
represented
by the
logic
of the
distributed
system
in
question
that
are
kind
of
ambiguous
or
rather
they're
manifest
in
classifiers
and
infomorphisms
that
aren't
defined
or

the
co-limit
the bold
c at the
top
there
does
not
exist
so
this
is
very
useful
criteria
I
think
to
distinguish
between
sort
of
empirical
models
which
the best
part
are
based
on
co-deployable
variables
and
what
happens
when
quantum
contextuality

occurs
that is
diagram
34
doesn't
commute
so
there's
some
sort
of
discrepancy
there
in the
mechanism
of the
quantum
reference
frame
relative
to the
dynamics
of
the
Markov
blanket
or
the
holographic
screen
underlying
it
so
there is
some
kind of
distortion
in

meaning
that
results
in this
sort
of
intrinsic
contextuality
I
don't know
if that's
a satisfactory
answer
at all
but
when
this
breaks
down
I
haven't
seen
that
before
I
I'm
sorry
there's
another
local
district
it's
all
good
Chris
did you
want to
add on

to that
or make
another
comment
well
I wanted
to go
back
a little
bit
to the
basics
of what
a quantum
reference
frame is
because
this is
a
concept
that
is
both
very
familiar
and I
think
a
little
bit
unfamiliar
and
the
term
was
only
coined
back

in the
1990s
when
the
idea
of
a
quantum
measurement
was
associated
with
the
idea
of
the
units
in which
the
measurement
is
made
and
what
are
the
requirements
for
a
measurement
being
meaningful
and we're
going to get
right back
to the
same issues
we've been

talking about
with respect
to time
and
thingness
but
a
reference
frame
is
just a
way of
giving
meaning
to an
outcome
and
it's
what
allows
outcomes
to be
compared
so if
I have
an
apparatus
like
a
meter
stick
I can
go around
and measure
lengths
and as
long as
I believe

that my
meter
stick
maintains
its
identity
as long
as I
believe
that my
meter
stick
remains
the
same
thing
then I
can
compare
the
lengths
that I've
measured
but if
I don't
believe
my
meter
stick
remains
the
same
thing
then
these
length
comparisons
are

meaningless

so

the

notion

of

a

reference

frame

is

intimately

tied

up

with

this

notion

of

meaning

which

in

this

free

energy

principle

context

is

really

a

notion

of

action

ability

can

I

use

what

something

means

to

guide
my
actions
in
my
decision
making
so
this
notion
of
a
QRF
becomes
a
notion
of
giving
of
giving
an
outcome
value
that I
obtain
and
write
in
memory
comparability
with
other
outcome
values
that I
maintain
and write
in memory

and
we're
very
used
to
you know
as
human
beings
we have
this
incredibly
complicated
sensory
and
inferential
apparatus
that
involves
many
many
degrees
of
freedom
and
many
many
values
that are
distinguishable
so I
think
if we
want
to
think
about
something

like
the
role
of
reference
frames
in
perception
it's
better
to
think
in
terms
of
something
like
E.
Coli
and
so
think
of a
much
simpler
kind
of
system
that
can
measure
fewer
things
right
E.
Coli
can
measure

concentrations
of salt
and
sugars
of
various
kinds
and
those
concentrations
are
meaningful
to
it
those
concentrations
allow it
to make
decisions
because
its
internal
biochemistry
assigns
them an
actionable
meaning
so
chemotaxis
and E.
Coli
works
because
the
level
of
phosphorylation
of a

particular
chemical
remains
reasonably
constant
that's
the
reference
frame
and
concentration
levels
can be
compared
to
each
other
as long
as that
phosphorylation
state
remains
reasonably
constant
that
reference
frame
is
meaningful
!
And so
chemotaxis
makes
sense
because
E.
Coli
has

if you
will
a
slow
clock
the
phosphorylation
state
of
Qi
that
confers
meanings
upon
its
outcomes
So
now we
can
spool
back
to
what
Jim
was
saying
about
co-deployable
reference
frames
reference
frames
that
I can
use
at the
same
time

that
I can
obtain
information
from
simultaneously
have to
commute
this
is
what
Heisenberg's
uncertainty
principle
is
about
and
if
I
can't
use
them
at
the
same
time
it's
because
they
don't
commute
if
I
use
them
in
different
orders

I
get
different
answers
and
what
that
means
is
that
by
switching
reference
frames
I'm
effectively
changing
the
calibration
I'm
effectively
changing
the
meaning
that
I've
assigned
and
I
think
a
very
beautiful
place
to
see
this
is

the
paper
that
we
referenced
here
from
Cervantes
and
Zafirov
called
the
Snow
Queen
Experiment
the
paper
is
called
Snow
Queen
is
Evil
and
Beautiful
came
out a
few
years
ago
and
basically
this
paper
is
demonstrating
quantum
contextuality

and
human
decision
making
and
it's
doing
it
by
exploiting
differences
context
dependent
differences
in the
meanings
of
words
so
it's
perfectly
pointing
out
this
relationship
between
meaning
and
co-deployability
and
hence
context
dependence
in
human
cognition
so
that's

a
paper
I
very
much
recommend
reading
thank
you
so
much
Carl
did
you
want
to
comment
on
that
also
I'm
aware
that
Stephen
also
had a
question
but
just
to
follow
up
on
that
beautiful
deconstruction
this is
more of a

question
but I'm
going to
state it
as
an
observation
this
notion
of
the
QRF
as being
fundamentally
getting
measurements
right
or metrics
right
would be
understood
from the
point of
view
of
inference
technically
as
getting
an internally
consistent
metric
correct
and the
metric
we're
talking
about

is a
Fisher
information
metric
so
if you
look at
all
reading
and
writing
or
action
and
sensation
from a
classical
perspective
as
just
trying
to
minimize
some
divergence
or
some
prediction
error
what you
are saying
is that
you are
trying
to
find
an
internally

consistent
metric
in an
information
space
that
makes
sense
of
your
active
sampling
querying
the
world
so
it
just
sounded
to
me
as
if
what
Chris
was
saying
was
exactly
getting
the
measuring
stick
at least
behaving
consistently
you'll never
know whether

it's changing
its length
or not
but at least
if it
behaves
consistently
in relation
to what
you can
see
of the
world
the way
that you
can
query
that
world
actively
then
that's
good
enough
and
that
of
course
is
one
way
of
simply
stating
the
name
of
the

game
in
active
inference
is
just
to
minimize
your
prediction
error
where
you're
now
reading
prediction
error
as
a
KL
divergence
and
of
course
the
information
length
or the
amount
the
measurement
of
movement
the
metric
is
just
a

path
integral
of
infinitesimal
changes
in KL
divergence
so
it's
seen
from a
mathematical
perspective
of
information
of
information
geometries
there's
a
beautiful
link
here
between
aligning
your
QRFs
and
getting
your
generative
models
right
to make
the most
sense
in the
sense

of
minimizing
!
or making
as
internally
consistent
as
possible
all your
information
metrics
awesome
thank you
before
we go
to
Steven's
question
we
have
a
question
or
two
from
the
from
and
in
the
live
chat
he
asks
does
the
existence

of
a
holographic
screen
slash
markup
blanket
automatically
yield
a
statistical
manifold
for
the
belief
state
of
the
agent
and
is
there
a
canonical
renormalization
procedure
for a
quantum
mechanical
system
if
there
is
one
how
would
the
core

screening
translate
into
the
statistical
manifold
I
can
take
one
quack
at
that
which
is
if
I
if
I'm
core
screening
something
if
we
think
of
this
in
terms
of
QRFs
then
I'm
deploying
a
different
reference
frame

with
lower
resolution
so
the
in a
sense
what's
being
renormalized
is
my
representation
of
the
units
that I
use
to
distinguish
states
of the
world
so
if
if
I
have
a
meter
stick
that's
made
of
stainless
steel
that
has

millimeters
indicated
on it
then I
can go
around and
make
measurements
in
millimeters
but if
I
coarse
grain
to a
meter
stick
that
only
has
half
centimeters
indicated
on it
then I've
lost a
factor of
five in
resolution
I've
coarse
grained
by a
factor
of five
and so
I've
in effect

rounded
off all
my
outcome
values
and
now
my
function
of
comparing
outcome
values
is
differently
defined
because
it has
to work
at this
lower
resolution
and
if I
want to
compare
my
previous
values
to my
later
values
I
have
to
incorporate
this
explicit

roundoff
function
into
my
comparison
of
values
because
I've
changed
my
QRF
so
so
coarse
graining
which
of course
couples
to scale
changing
from the
very
beginning
of our
conversation
can be
thought of
as a
transition
from
a
very
a
QRF
with some
grain size
to a

QRF for a
similar degree
of freedom
with a
different grain
size
and that
has ripple
effects
through
everything
including
the
free
energy
cost
of
memory
right
if I'm
writing
2-bit
numbers
into
memory
then that's
a lot
less expensive
than writing
5-bit
numbers
into
memory
so
this
raises
another
aspect

of this
question
of
renormalization
which is
how does
it
influence
how can
I use
it
to
solve
the
trade-off
problems
that arise
if I'm
an organism
with a
limited
fuel supply
which of
course I
am
going
about
the
world
trying
to
gather
preferentially
gather
the
information
that's
going to

be
most
useful
to
me
at
the
lowest
possible
cost
so
again
we get
back
to
this
key
issue
in
all of
active
inference
of
balancing
of
the
risk
of
exploration
against
the
reward
of
new
information
and
course
graining

is
intimate
to
that
and
it
involves
the
choice
I
make
of
the
tools
that
I'm
going
to
use
to
make
measurements
and
how
that
choice
impacts
my
need
to
allocate
computational
resources
and energy
resources
to what
I'm
doing

to
the
complex
task
of
getting
new
information
while
at the
same
time
surviving
awesome
thanks
Daniel
did you
oh
Carl
do you
want to
speak
to that
first
perhaps
Steven
should
ask
to
what
just
said
yeah
I'll
make
a
note
and

then
Carl
it'd
be
great
to
have
your
response
and
then
Steven
to
your
question
so
to
transpose
it
into
statistical
parametric
mapping
the
true
classics
there's
a
mapping
from
the
measured
voxels
which
are
being
defined
by

the
resolution
of
the
EEG
or
the
fMRI
machine
that's
the
holograph
that's
like
what
the

experimental
apparatus
is
providing
is
the
EG
and
that
space
you know
if it's
128
channels
in the
EEG
and
they're
getting
this
sampling

right
that
is
the
state
space
that's
being
discussed
and
that's
being
transposed
or mapped
or
connected
to a
different
statistical
space
with
different
features
and
rather
than
the
voxels
there's
like
the
mapping
onto
res
else
so
Carl
that's

why
I'd
appreciate
if
this
is
true
or
not
but
those
are
the
resolution
elements
and
that
it
can
be
like
a
coarse
graining
and
in
that
stretch
meter
stick
and
different
space
there's
a lot
more
statistical
inference

options
available
that are
just
not
accessible
at
all
from
the
voxel
coarse
graining
which is
noisier
and
there's
a signal
enrichment
in the
coarse
graining
to
the
res
that's
appropriate
and then
that gets
juxtaposed
with the
contrasts
which is
the
experimental
design
of the
scientist

like you
couldn't
differentiate
A versus
B
category
if you
only
tested
A
so the
experimental
design
that's where
the statistics
actually comes
into play
and then
and then
just as
Chris said
there's
the
organism
with
limited
food
supply
and
then
there's
like
the
scientist
with
limited
time
attention

computer
storage
processor
time
funding
maybe
7 billion
people
like
there's
plausibility
and so
that's
where this
operational
co-deployment
connects with
experimental
design
and so
many of
the
patterns
and
analogies
exist in
the purely
classical
domain
of the
parametric
statistics
like SPM
in the
classical
physics
and the
flow

and the
information
flow
area
that's
coming
into play
and also
these
more
recent
in
quantum
areas
so
Carl
would love
to hear
your
thoughts
on that
yes
of course
I'm
mindful
that
Stephen
still
got his
hand
down
but
perhaps
I'll
speak
to
that
and

then
Stephen
can
ask
the
question
but
I
think
these
questions
and
Daniel
your
responses
are
touching
on
something
really
fundamental
so
just
to
answer
the
question
from
the
point
of
view
of
Markov
Blanket
yes
it
does

automatically

yield

a

statistical

manifold

or

at

least

it

looks

as

if

there

is

a

statistical

manifold

if

you

can

observe

the

behaviour

of

the

thing

that

you

are

observing

the

second

thing

question

which

we've

been

focusing

on
the
coarse
graining
I
think
that's
a
really
fundamental
thing
I
would
argue
though
not
to
conflate
too
much
the
notion
of
renormalisation
which
would
be
I
think
more
a question
of how
you get
from
one
scale
to
another

scale
with
the
notion
of
coarse
graining
which
could be
read
just
simply
as
the
degree
of
quantisation
or
choosing
the
right
resolution
element
the
right
coarse
graining
of a
given
QRF
at
any
given
scale
you
could
argue
that

you
could
contextualise
it
but
I
think
there's
a
more
fundamental
argument
here
and
that's
the
argument
that
there
will
be
from
a
statistical
perspective
an
optimal
coarse
graining
for
any
given
holographic
screen
or
exchange
over
a

Markov
blanket
and
that's
the
one
that
statistically
maximises
the
marginal
likelihood
which
you
could
if
you
like
relate
to
this
sort
of
getting
the
measurements
as
internally
consistent
as
possible
maximising
the
predictability
of the
next
measurement
you

make
that
goes
hand
in
hand
with
a
maximisation
of
the
marginal
likelihood
or
minimisation
of
the
variation
free
energy
that
is
important
to
remember
because
the
marginal
likelihood
can
always
be
expressed
as
accuracy
minus
complexity
where

complexity
scores
the
degrees
of
freedom
you're
burning
up
if you
remember
about
the
bits
on the
holographic
screen
you need
energy
to
from a
classical
perspective
represent
and
what
that
translates
to
in a
very
simple
way
in
terms
of
probability
distributions

and
functionals
is that
in maximising
the marginal

likelihood
or the
evidence
for your
aligned
or
model of
the world
complexity
you are
to
maintain
a
certain
degree
of
accuracy
trying
to
minimise
complexity
so
you're
using
as
few
degrees
of
freedom
you're
coarse
grading

to
the
extent
you
can
get
away
with
while
maintaining
an
accurate
account
of
the
of
say
the
sensory
states
or
the
perceptual
space
on your
holographic
screen
if
you
know
from
the
point
of
view
of
active
inference

just
think
about
what
would
that
sort
of
accuracy
and
complexity
look
like
in
terms
of
the
consequences
of
action
you
get
exactly
what
Chris
was
talking
about
which
was
the
expected
complexity
being
the
risk
usually
and I

mean
risk
in the
sense
of
an
economist
or
decision
theorist
it
is
literally
mathematically
the
same
thing

The
expected
accuracy
now
becomes
your
sort
of
expected
utility
or
the
expected
log
evidence
the
likelihood
the
expected
likelihood

that you
would
get
if you
committed
to this
kind
of
coarse
graining
so
this
coarse
graining
is
absolutely
fundamental
it's
really
interesting
to
go
back
to
all
the
sources
where
the
coarse
graining
starts
to
make
a
difference
and
I'm

thinking
here
coming
back
to
Daniel's
notion
of
resolution
elements
you
know
you
find
this
in
sort
of
Vena
filtering
you
find
the
match
filtering
theorem
there
is
an
optimal
level
of
smoothing
or
blurring
for
any
given

kind
of
data
about
which
you
want
to
make
an
inference
or
you
want
to
assimilate
you
see
this
in
formulations
of
universal
computation
that
themselves
are
based
upon
Solomov
induction
that
get
you
right
back
to
complexity

and
compressibility
so
you
see
it
in
the
drive
that
underwrites
all
formulations
of
universal
computation
to
maximise
compression
by
minimising
the
complexity
of your
message
passing
getting
as
efficient
and
as
simple
as
you
can
and
of
course

the
best
way
to
make
your
account
as
simple
as
possible
is
to
course
grade
it
and
I
think
you
also
see
it
in
the
different
levels
of
cognition
in
psychology
and
in
neurophysiology
in the
sense
that
the

higher
level
more
efficient
ways
in which
we
align
our
QRFs
or
optimise
our
generative
models
always
lead
you
to
a
very
course
grade
quantisation
of the
world
so
for
example
I
represent
I'm
sure
in
my
brain
on
some

neural
populations
the
fact
that
I
am
in
my
study
I
do
not
represent
that
in
terms
of
very
fine
grained
x y
z
coordinates
down
to
the
millimetre
I
represented
no
I'm
in
my
study
I
carve
the

world
at
its
joints
and
those
joints
basically
demarcate
the
kind
of course
training
I
bring
to
the
table
there
are
all
sorts
of
arguments
you
can
pursue
here
to
vindicate
this
the
very
existence
of
receptive
fields
tiling

a
sensorium
in a
way
that is
covered
by
discrete
quanticized
receptive
fields
tells you
immediately
that
the
way
the
brain
represents
stuff
on
the
inside
if
you
can
look
at
it
through
brain
imaging
is
in
this
quanticized
fashion
this

coarse
grained
fashion
and
that
coarse
grained
ness
and
the
size
and
the
deployment
and
the
organisation
of
those
receptive
fields
has
been
optimised
at
many
different
temporal
scales
!
generative
model
or
to
the
way
that
we

align
our
QRFs
to
make
sense
of
the
stuff
on
the
holographic
screen
or at
least
when
we
are
reading
from
that
sense
making
in
that
sense
Great
thank you
Stephen
did anyone
ask a
question
Yeah
thanks
no
this is
really
cool

I was
going to
come
back
a
little
bit
to
now
in
the
past
and
the
idea
that
you
mentioned
about
the
way
that
we
interface
or
the
assumptions
of
the
interface
then
changes
how
we
resolve
the
thing
and

the
background
situation
situatedness
of that
thing
so
that
could
then
also
and
this
is
the
question
in
a
way
is
how
can
that
also
because
if
the
way
that
thing
and
the
assumptions
are
made
sets
up
the

meaningful

influence

the

action

policy

selection

and the

way

things

are

thought

about

how

can

that

itself

change

the

way

the

brain

or the

way

the

organisms

look at

what

things

are

for

instance

in

the

modern

world

we

do

see

everything
as
things
we
carve
things
up
we
see
things
as
separate
like
an
individual
is
separate
whereas
if
you
go
back
to
indigenous
approaches
time
for
them
was
cyclical
or
followed
the
river
going
down
a
hill

or
so
there
was
a
complexity
focus
in a
way
it was
almost
like
the
mountain
car
problem
of
finding
the
landscape
of
meaning
rather
than
defining
the
thing
and
what
it
is
and
carving
it
so
I'm
curious
around

how
much
is
the
way
the
quantum
reference
field
translated
into
thinking
about
what
that
thing
is
and
how
that
compares
to
our
western
cognitive
frames
which
we measure
everything
because
we can
because
we can
control
our
environment
so
heavily

we can
make
things
very
accurate
whereas
in the
past
it
might
have
been
the
Kalahari
Bushmen
or
people
in
the
Amazon
that
somehow
go
with
the
landscape
of
complexity
so
that
the
way
that
that
is
understood
is
going

to
be
different
so
I'm
just
curious
about
how
you
think
about
that
ways
of
knowing
!
to
the
temporal
scales
like you
mentioned
the
cyclical
time
time
and
back to
Franz's
question
earlier
about
how
a
thing
over
time

that
stays
a
thing
over
time
scales
with
the
scale
of
the
time
or
the
cyclical
nature
of
the
time
or
it
reminds
me
back
of
when
we
had
Chris
and
Shauna
here
talking
about
flattening
time
maybe

that
was
just
Shauna's
talk
but
yeah
anyway
I don't
know
do you
guys
anybody
have
any
temporal
comments
just
that
there's
an
analogy
to
the
synchronization
of
clocks
which
is a
huge
part
of
computer
science
and
of
quantum
and

other
areas
that
question
of
synchronizing
clocks
seems
to have
some
relationship
to
cognitive
model
not
naive
congruence
in terms
of
synchronization
but
in
terms
of
time
zone
coordination
so
if
we're
in
the
same
time
zone
then
there's
a

comparability

but

for

cognitive

states

and

so

I

think

that

will

be

something

awesome

to

explore

in

next

week's

session

and

of

course

in

the

long

run

about

what

that

kind

of

semantic

and

rhetorical

and

narrative

alignment

that
isn't
congruent
but
reflects
a
broader
frame
which
actually
the
differences
within
the
frame
are
the
search
through
that
field
thanks
and
in
the
interest
of
temporal
comments
I
think
maybe
does
everyone
just
want
to
go

around
and
give
a
final
thought
and
maybe
an
idea
of
what
they
would
like
to
see
in
next
week's
discussion
and
we
will
start
with
Stephen
yeah
thanks
lots
of
interesting
things
I
suppose
I'm
curious
about

this
the
way
that
collapsing
of
the
quantum
reference
frame
might
scale
differently
or
might
the
assumptions
that
lead
to
the
collapse
could
inform
different
scales
of
interactions
and
mixing
and
blending
of
behavior
and
yeah
digesting
a lot

of stuff
that's
come up
so
yeah
thank
you
for
the
opportunity
to
talk
to
you
awesome
daniel
anything
you
want
to
see
happen
next
week
we
always
think
of
the
dot
zero
it's
kind
of
like
dropping
into
the

skate
park
or
the
bathtub
and
so
that
was
like
a
microcosm
of
all
the
threads
that
get
brought
together
and
then
in
dot
one
I
feel
like
we
opened
up
so
many
areas
of
possible
connection
and

so
I'll
just
look
forward
to
the
second
repeated
measurement
of
this
semantic
or
regime
of
attention
space
and
see
how
the
continued
interaction
and
engagement
updates
our
generative
model
past
the
dot
to
even
awesome
Carl
or Chris

do you
want to
give a
thought
about
what
you'd
like
to
unpack
in
the
next
question
should I
go first
well
Chris
thinks
of
something
very
clever
to
say
so
it's
following
on
from
Daniel
to
certain
extent
Stephen's
question
as
well

I'd
like
to
hear
about
the
principle
of
unitarity
entanglement
and
its
relationship
to
this
notion
of
synchronization
and
the
mutual
alignment
of
QRFs
between
things
that
are
coupled
so
coming
to
the
regimes
of
attention
I
think

that
could
be
a
very
transparent
link
there
in
the
sense
that
contextualizing
your
QRF
I
think
for a
cognitive
neuroscientist
would be
getting
your
attentional
set
right
and
possibly
implicit
coarse
graining
that
comes
along
with
a
particular
attentional

set
taking
that
notion
to
tending
to
others
and
synchronizing
and
communicating
and living
in a
universe
that is
also
composed
of
things
like
me
brings
some
fundamental
questions
about
classically
generalized
synchronization
quantum
theoretically
entanglement
and
enabling
that
entanglement
in the

right
kind
of
way
via
aligning
your
QRS
and
getting
the
course
graining
right
in
sympathy
with
your
partner
so
I'd
like
to
hear
about
the
principal
unitarity
entanglement
and how
it
relates
to
classical
notions
of
communication
mutual

inference
and
generalized
synchrony
well
thanks
for
stealing
what
I
was
going
to
say
I
was
also
going
to
say
how
there's
one
like
the
final
concluding
part
of
the
paper
talks
about
how
the
FEP
is
asymptotically

the
principle
of
unitarity
and
I
really
would
like
to
dive
into
that
more
and
maybe
any
potential
implications
of
that
and
now
Chris
gets
the
final
and
cleverest
word
okay
this
won't
be
clever
what
I
was

!

going

to

do

was

to

recommend

a

lovely

paper

by

Alexei

Grinbaum

that

we

refer

to

in

this

paper

and

the

closing

sentence

of that

paper

is that

physics

is really

about

languages

and

I think

this ties

in nicely

to some

of the

issues
that
Stephen
was
raising
in his
last
question
which
I
found
very
interesting
about
how
do
conceptual
schemes
evolve
at the
cultural
level
and
in
hearing
about
the
whole
issue
of
reference
frame
synchronization
or
reference
frame
alignment
and how

that
relates
to
entanglement
this
really
is a
question
about
language
and
that
paper
by
Alexei
Grinbaum
I think
makes
this
point
very
beautifully
so
that
would
be
a
nice
thing
to
look
at
at
some
point
let's
see
this

is
that's
probably
a
newer
paper
by
him
the
one
I
was
referring
to
is
titled
something
like
how
device
free
how
device
independent
methods
change
the
nature
of
physical
theory
or
something
like
that
was
published
in

studies
in
history
and
philosophy
of
science
2017
or
something
like
that
okay
anyway
thank you
very much
for
this
yeah
this
was
very
interesting
and
engaging
and
really
appreciated
everyone's
participation
and
blues
facilitation
so
we'll
look
forward
to

seeing

anyone

just

drop

in

or

out

when

works

for

you

next

week

at

the

same

time

in

number

40.2

so

till the

next

measurement

thank you

take care

to

Thank you.